

Ayush Priyadarshi

linkedin.com/in/ayush-priyadarshi | ayush729p.github.io | ayush729p@gmail.com | (623) 213-0379

EDUCATION

Master of Science in Mechanical Engineering

Expected May 2025

Arizona State University

Tempe, AZ, USA

- CGPA: 4.00/4.00
- Elected to Tau Beta Pi AZ B Chapter for outstanding academic performance
- Relevant course work: Advanced Material Characterization, Fundamentals of Semiconductor Packaging, Design of Engineering Experiments, Stress Analysis, Mechanics of BioMEMS and Biomaterials, Polymers and Composites

Bachelor of Technology in Mechanical Engineering

May 2023

Indian Institute of Technology (IIT), Ropar

Punjab, India

- CGPA: 8.14/10.00
- Relevant course work: Additive Manufacturing, Analysis of Material Removal Processes, Manufacturing Technology - I & II, Manufacturing Lab, Engineering Workshop, Industrial Management, Data Structures, Calculus

TECHNICAL SKILLS

- Design and Manufacturing: SolidWorks, ANSYS, ABAQUS, AutoCAD, 3D Printing (FDM, PBF), CNC Programming
- Software: MATLAB, C++, Python, SQL, Excel (with Macros), JMP, Origin Pro
- Engineering Practices: Total Productive Maintenance (TPM), Product Life Cycle Assessment (LCA), Ishikawa Diagrams
- Certifications: Certified SolidWorks Associate (CSWA), MATLAB Onramp (incl. Simulink & Simscape)

PROFESSIONAL EXPERIENCE

Research Assistant

June 2024 - Present

Bio & Nano Lab; Dr. Wonmo Kang

Tempe, AZ, USA

- Manufacture auxetic structures operating FDM 3D printers with ABS, PP, TPU, and PBF metal printing.
- Conduct compression and tensile testing for quasi-static analysis; validate results with FEA models.
- Optimize geometry and print parameters, improving performance of brittle and ductile materials for energy absorption.

Teaching Aide

July 2024 - Present

School for Engineering of Matter, Transport & Energy

Tempe, AZ, USA

- Teach and demonstrate structural mechanics lab experiments; guide students through bending, torsion, and vibration testing.
- Streamline course delivery by integrating Canvas and Ed Discussion to resolve student queries efficiently.
- Highlight importance of material testing in structural applications, emphasizing lab safety and accuracy.

Manufacturing Intern

June 2022 - August 2022

Trident Group Ltd

Dhaura, Punjab, India

- Led a critical analysis project resulting in a 15% reduction in production delays by optimizing Cross Cutting Cross Hemming (CCCH) machines' efficiency through detailed root cause analysis and process improvements.
- Managed a team of 8 field technicians as a shift engineer, enabling real-time identification of bottlenecks and implementation of process improvements at a leading textile manufacturer.
- Enhanced quality control and operational efficiency using Total Productive Maintenance (TPM) methodologies including Ishikawa diagrams and the 5 Whys approach.

PROJECT EXPERIENCE

Optimization of Keyboard Typing Efficiency

February 2024 - April 2024

- Applied ergonomic research deploying a 2^3 factorial design to analyze impact of keyboard type, wrist posture, and tilt angle on typing speed and accuracy, using custom test scores for normalized performance comparison.
- Conducted controlled experiments with randomized runs on mechanical and membrane keyboards, identifying mechanical keyboards and higher tilt angles as significant factors in improving typing efficiency.

Development of Auxetic Structures for defense applications

August 2022 - April 2023

- Designed 3D geometries of different cell shapes (with negative Poisson's ratio) and simulated against boundary conditions of static loading for performance analysis leveraging SolidWorks and ANSYS.
- Performed Impact analysis of bullet impact and parametric study of cell size dimensions for optimum shape and size of unit cells yielding 1.5x better properties.

Automatic mold making technique for instant forming

January 2023 - May 2023

- Devised a prototype mold for plastic forming processes (extrusion); to change profile on demand using SolidWorks.
- Developed a python and an arduino program to automate profile creation from 3D geometry data and motion of underlying mechanism.

PROFESSIONAL & CAMPUS INVOLVEMENT

CIM Club, IIT Ropar, *Mentor/Representative*

November 2020 - October 2022

- Led a community of over 100+ CAD/Simulation enthusiasts, fostering an innovative learning environment and enhancing collaborative skills, mirroring ethos of leading-edge manufacturing firms.
- Conducted numerous hands-on sessions on 3D design software, including SOLIDWORKS, ANSYS, and AutoCAD